

(1) You have to complete the cheatsheet in your own handwriting in coloured ink – no black ink may be used. (2) No problems or solutions to problems or examples from a textbook may be written on this cheatsheet. (3) No derivation of equations may be written on this cheatsheet. (4) Do not fold this cheatsheet; a folded cheatsheet will be confiscated. (5) You have to hand in your cheatsheet with your test – we will subtract 10% if there is no cheatsheet with your test. (6) Complete your cheatsheet in the space left below AND on the back side of the page (A4 paper size). (7) You may only have ONE cheatsheet in the exam venue.

Kinematics (const a)

$$v_f = v_i + at$$

$$x_f = x_i + v_i t + \frac{1}{2} at^2$$

$$v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$x_f = x_i + \frac{1}{2}(v_i + v_f)t$$

Projectile

$$v_{xi} = v_i \cos \theta$$

$$v_{yi} = v_i \sin \theta$$

(Choose $x_i = 0$ & $y_i = 0$)

$$x = v_{xi} t$$

$$y = v_{yi} t - \frac{1}{2} at^2$$

$$y = \left(\frac{v_{xi}}{v_{yi}}\right)x - \left(\frac{g}{2v_{yi}^2}\right)x^2$$

$$t_H = \frac{y}{v_{yi}} = \frac{v_{yi} \sin \theta}{g}$$

$$v_{yH} = v_{yi} - gt_H = v_{yi} \cos \theta = \frac{v_i^2 \sin 2\theta}{2g}$$

$$t_R = 2t_H$$

$$x_R = R = v_{xi} t_R = \frac{v_i^2 \sin 2\theta}{g}$$

Relativity

$$\vec{r}_{PA} = \vec{r}_{PB} + \vec{v}_{BA} t$$

$$\vec{v}_{PA} = \vec{v}_{PB} + \vec{v}_{BA}$$

$$\vec{a}_{PA} = \vec{a}_{PB}$$

Newton

- $\vec{F}_{net} = \sum \vec{F}_i = 0$
- $\vec{F}_{net} = \sum \vec{F}_i = m\vec{a}$
- $\vec{F}_{2,1} = -\vec{F}_{1,2}$

$$W \quad N \quad F \quad T \quad F_s \quad F_d \quad F_{ext}$$

Uniform circular motion

$$a_c = \frac{v^2}{r} = r\omega^2$$

$$T = \frac{2\pi r}{v}$$

$$\omega = \frac{2\pi}{T} = \frac{v}{r}$$

$$\sum F_r = ma_c = m \frac{v^2}{r}$$

Non-uniform circular motion

$$\vec{a} = \vec{a}_c + \vec{a}_t$$

$$\sum \vec{F} = \sum \vec{F}_r + \sum \vec{F}_t$$

$$T = m\omega^2 r (\cos \theta + \cos \phi)$$

Energy

$$W = F \cos \theta \quad (\text{const } F)$$

$$W_{ext} = \int_{x_i}^{x_f} F(x) dx$$

$$W = F \Delta x$$

$$W_{ext} = \sum W = \int (\sum F) \cdot dx$$

$$F_s = -kx = -F_{ext}$$

$$W_s = \frac{1}{2} kx_f^2 - \frac{1}{2} kx_i^2$$

$$W_s = \int_{x_i}^{x_f} (-kx) dx = -W_{ext}$$

KE

$$W_{ext} = \Delta K$$

$$K = \frac{1}{2} mV^2$$

$$K_f = K_i + W_{ext}$$

PE

$$W_{ext} = \Delta U$$

$$U_{grav} = mgy$$

$$U_{spring} = \frac{1}{2} kx^2$$

Conservative forces

$$W_{con} = -\Delta U = W_{int}$$

$$E_{mech} = 0$$

$$F_{oc} = -\frac{dU(x)}{dx}$$

Non-conservative forces

$$E_{mech} = K + U$$

Grav PE

$$F_{grav}(r) = -\frac{GMm}{r^2}$$

$$U_{grav}(r) = -\frac{GMm}{r}$$

Non-isolated

$$\Delta E_{system} = \sum E = \Delta K + \Delta U + \Delta E_{int}$$

$$W = \Delta K$$

Isolated

$$\Delta K + \Delta U = 0$$

$$\Delta E_{mech} = 0$$

$$\Delta E_{system} = 0$$

$$E_{mech,i} = E_{mech,f}$$

Kinetic Friction

$$\sum W_{fric} = F_k d = \Delta K$$

$$\Delta E_{int} = f_k d$$

$$\Delta E_{mech} = \Delta K + \Delta U_{grav} = -f_k d$$

$$\Delta K + \Delta U + \Delta E_{int} = 0$$

Power

$$P_{avg} = \frac{W}{\Delta t}$$

$$P = \frac{dW}{dt} = \vec{F} \cdot \vec{v}$$

Linear momentum (isolated)

$$\sum \vec{p} (m_i \vec{v}_i + m_i \vec{v}_i) = 0$$

$$\vec{p} = m\vec{v}$$

$$\vec{p}_{tot} = \vec{p}_i + \vec{p}_f$$

$$\Delta \vec{p}_{tot} = 0$$

$$\vec{p}_{tot,i} = \vec{p}_{tot,f}$$

$$\sum \vec{F} = \frac{d\vec{p}}{dt}$$

Momentum non-isolated

$$\Delta \vec{p} = \int_{t_i}^{t_f} \sum \vec{F} dt = \vec{I}$$

$$\vec{I} = \sum \vec{F} \Delta t \quad (\text{const } \sum \vec{F})$$

Inelastic collision

$$\Delta \vec{p} = 0$$

$$\vec{v}_f = \frac{m_1 \vec{v}_i + m_2 \vec{v}_i}{m_1 + m_2}$$

$$\vec{v}_f = \frac{m_1}{m_1 + m_2} \vec{v}_i \quad (v_{2i} = 0)$$

Elastic collision

$$\Delta \vec{p} = 0 \quad \& \quad \Delta K = 0$$

$$v_{if} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) v_{ii} + \left(\frac{2m_2}{m_1 + m_2}\right) v_{2i}$$

$$v_{2f} = \left(\frac{2m_1}{m_1 + m_2}\right) v_{1i} + \left(\frac{m_2 - m_1}{m_1 + m_2}\right) v_{2i}$$

$$v_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) v_{1i} \quad (v_{2i} = 0)$$

$$v_{2f} = \left(\frac{2m_1}{m_1 + m_2}\right) v_{1i} \quad (v_{2i} = 0)$$

Cent

$$r_{cm} = \frac{1}{M} \sum m_i r_i$$

$$dI = \sum m_i r_i^2$$

Many particles

$$V_{cm} = \frac{d\vec{r}_{cm}}{dt} = \frac{1}{M} \sum m_i \vec{v}_i$$

$$\vec{p}_{tot} = \sum \vec{p}_i = \sum m_i \vec{v}_i$$

$$\vec{a}_{cm} = \frac{d\vec{v}_{cm}}{dt} = \frac{1}{M} \sum m_i \vec{a}_i$$

$$\sum \vec{F}_{ext} = M \vec{a}_{cm}$$

RRO

$$s = r\theta$$

$$V = r\omega$$

$$a_t = r\alpha$$

$$W_{ang} = \frac{\Delta \theta}{\Delta t} \quad ; \quad \omega = \frac{d\theta}{dt}$$

$$L_{ang} = \frac{\Delta W}{\Delta t} \quad ; \quad \alpha = \frac{d\omega}{dt}$$

Torque

$$d = r \sin \phi$$

$$\tau = r F \sin \phi = Fd$$

$$\vec{\tau} = \vec{r} \times \vec{F} \quad (\text{S})$$

$$\vec{\tau} > 0 \text{ (ccw)} \quad ; \quad \vec{\tau} < 0 \text{ (cw)}$$

RRO under torque

$$\sum \vec{\tau}_i = m\vec{a}_{cm}$$

$$\sum \tau = \tau_{net} = (\sum F_i) r = I\alpha$$

$$I = \int r^2 dm = mr^2$$

RRO

$$W_f = W_i + \alpha t$$

$$\theta_f = \theta_i + \omega_i t + \frac{1}{2} \alpha t^2$$

$$\omega_f^2 = \omega_i^2 + 2\alpha(\theta_f - \theta_i)$$

$$\theta_f = \theta_i + \frac{1}{2}(\omega_i + \omega_f)t$$

<u>Rotational KE</u> $K_r = \sum K_i = \frac{1}{2} I \omega^2$ <u>Energy cons. RRO</u> $ ds = r d\theta$ $dW = \vec{F} \cdot d\vec{s}$ $dW = \tau d\theta$ $P = \tau \omega$ $W_{\text{ext}} = \Delta K_r$	$y = A \sin(kx - \omega t + \phi)$ $\vec{v} = \lambda f$ $v_{\text{max}} = \omega A$ $a_{\text{max}} = \omega^2 A$ <u>Superposition</u> $y(x,t) = z A \cos(\frac{\theta}{2}) \sin(kx - \omega t + \frac{\phi}{2})$ Constructive $\phi = \text{even multiple } \pi$ Interference $\phi = \text{odd multiple } \pi$	<u>Elements of Inertia (I_{cm})</u> Thin cylinder: $\frac{1}{2} MR^2$ Hollow cylinder: $\frac{1}{2} M(R_1^2 + R_2^2)$ Solid cylinder: $\frac{1}{2} MR^2$ Rectangular plate: $\frac{1}{12} M(a^2 + b^2)$
<u>SHO spring</u> $F_s = -kx$ $a_x = -\frac{k}{m}x$ $\frac{d^2x}{dt^2} = -\omega^2 x$ $\omega = \sqrt{\frac{k}{m}}$ $x(t) = A \cos(\omega t + \phi)$ $v(t) = -A\omega \sin(\omega t + \phi)$ $a(t) = -A\omega^2 \cos(\omega t + \phi)$ $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$ $f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$ $v_{\text{max}} = \omega A$ $a_{\text{max}} = \omega^2 A$	<u>Standing waves</u> $y(x,t) = z A \sin(kx) \cos(\omega t)$ Nodes: $\sin(kx) = 0$ $x = \frac{n\lambda}{2}, n=0,1,2,\dots$ Anti-nodes: $\sin(kx) = \pm 1$ $x = \frac{n\lambda}{4}, n=1,3,5,\dots$	Rod center: $\frac{1}{12} ML^2$ Rod end: $\frac{1}{3} ML^2$ Solid sphere: $\frac{2}{5} MR^2$ Hollow sphere: $\frac{2}{3} MR^2$
<u>Energy SHO</u> $\Delta E_{\text{mech}} = 0$ $E_{\text{mech}} = K + U$ $K = \frac{1}{2} k A^2 \sin^2(\omega t + \phi)$ $U_{\text{spring}} = \frac{1}{2} k A^2 \cos^2(\omega t + \phi)$ $E_{\text{mech}} = \frac{1}{2} k A^2$	<u>Temperature</u> $T_c = T - 273.15$ $\alpha = \frac{\Delta L / L_i}{\Delta T}$ $\Delta L = \alpha L_i \Delta T$ $\Delta V = \beta V_i \Delta T \quad (\beta = 3\alpha)$ <u>1st law thermodynamics</u> $Q = mc \Delta T$ $T_c < T_f < T_h$ $-Q_{\text{hot}} = +Q_{\text{cold}}$	
<u>Pendulum (simple)</u> $\sum F_t = mg_t$ $a_t = \frac{dv_t}{dt}$ $s = L\theta$ $\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta \quad (\theta < 10^\circ)$ $\frac{d^2s}{dt^2} = -\omega^2 s$ $\omega^2 = \frac{g}{L}$ $\theta = \theta_{\text{max}} \cos(\omega t + \phi)$	<u>Latent heat</u> $Q = L \Delta m$ <u>Extra</u> $\rho = \frac{m}{V}$ (Density) $v_{\text{avg}} = \frac{\Delta x}{\Delta t}; V = \frac{\Delta V}{\Delta t}$ $a_{\text{avg}} = \frac{\Delta v}{\Delta t}; a = \frac{dv}{dt}$ $\vec{A} \parallel \vec{B} \Rightarrow \vec{A} \times \vec{B} = 0$ $\vec{A} \perp \vec{B} \Rightarrow \vec{A} \times \vec{B} = AB$ $\vec{A} \times \vec{B} = AB \sin \theta$ $\frac{d}{dt}(\vec{A} \times \vec{B}) = \frac{d\vec{A}}{dt} \times \vec{B} + \vec{A} \times \frac{d\vec{B}}{dt}$ $\vec{A} \cdot \vec{B} = AB \cos \theta = A_x B_x + A_y B_y$	
<u>Pendulum (physical)</u> $\tau = -mgL \sin \theta$ $\frac{d^2\theta}{dt^2} = -\frac{mg}{I} \sin \theta = -\omega^2 \theta$ $\omega = \sqrt{\frac{mg}{I}}$	<u>Wave motion</u> $y(x,t) = f(x - Vt)$ $y(x,t) = A \sin[\frac{2\pi}{\lambda}(x - Vt)]$ $\lambda = \frac{2\pi}{k}$ $\omega = \frac{2\pi}{T} = 2\pi f$	